



MATLAB Drive > Dsp2_1.m

```
1 clc;
2 clear;
3 close all;
4
5 % Specifications
6 Wp = 0.4;
7 Ws = 0.55;
8 Rp = 1;
9 Rs = 40;
10
11 % Filter Order
12 [N,Wn] = cheblord(Wp,Ws,Rp,Rs);
13
14 % Design LPF
15 [b,a] = cheby1(N,Rp,Wn,'low');
16
17 fprintf('Filter Order = %d\n',N);
18 fprintf('Cutoff Frequency = %.2f*pi rad/sample\n',Wn);
19
20 % Stability Check
21 poles = roots(a);
22
23 if all(abs(poles)<1)
24     disp('System is STABLE');
25 else
26     disp('System is UNSTABLE');
27 end
28
29 % Magnitude and Phase Response
30 figure;
31 freqz(b,a);
32 title('Chebyshev Type-I Low Pass Filter');
33
34 % Impulse Response
35 figure;
36 impz(b,a);
37 title('Impulse Response');
38
39 % Group Delay
40 figure;
41 grpdelay(b,a);
42 title('Group Delay');
43
44 % Pole Zero Plot
45 figure;
46 zplane(b,a);
47 title('Pole Zero Plot');
```



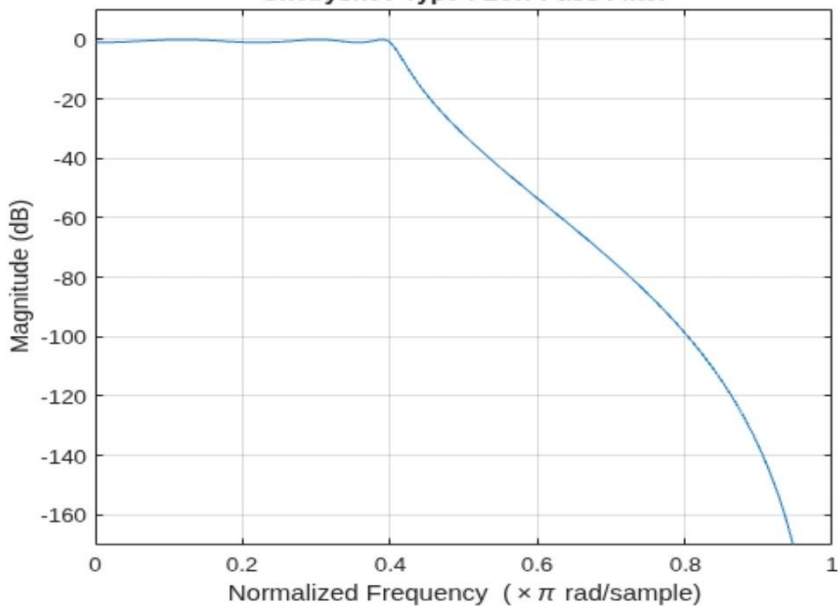
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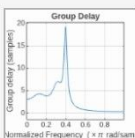
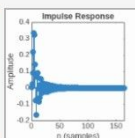
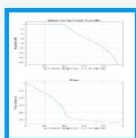
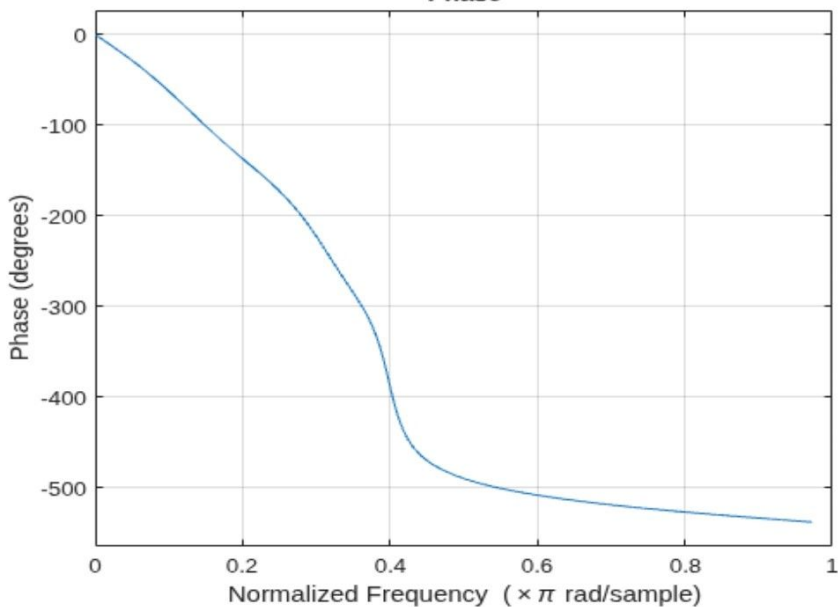


Figure 1: Dsp2_1

Chebyshev Type-I Low Pass Filter



Phase



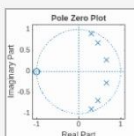
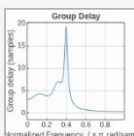
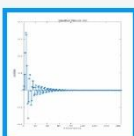
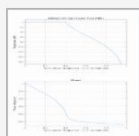
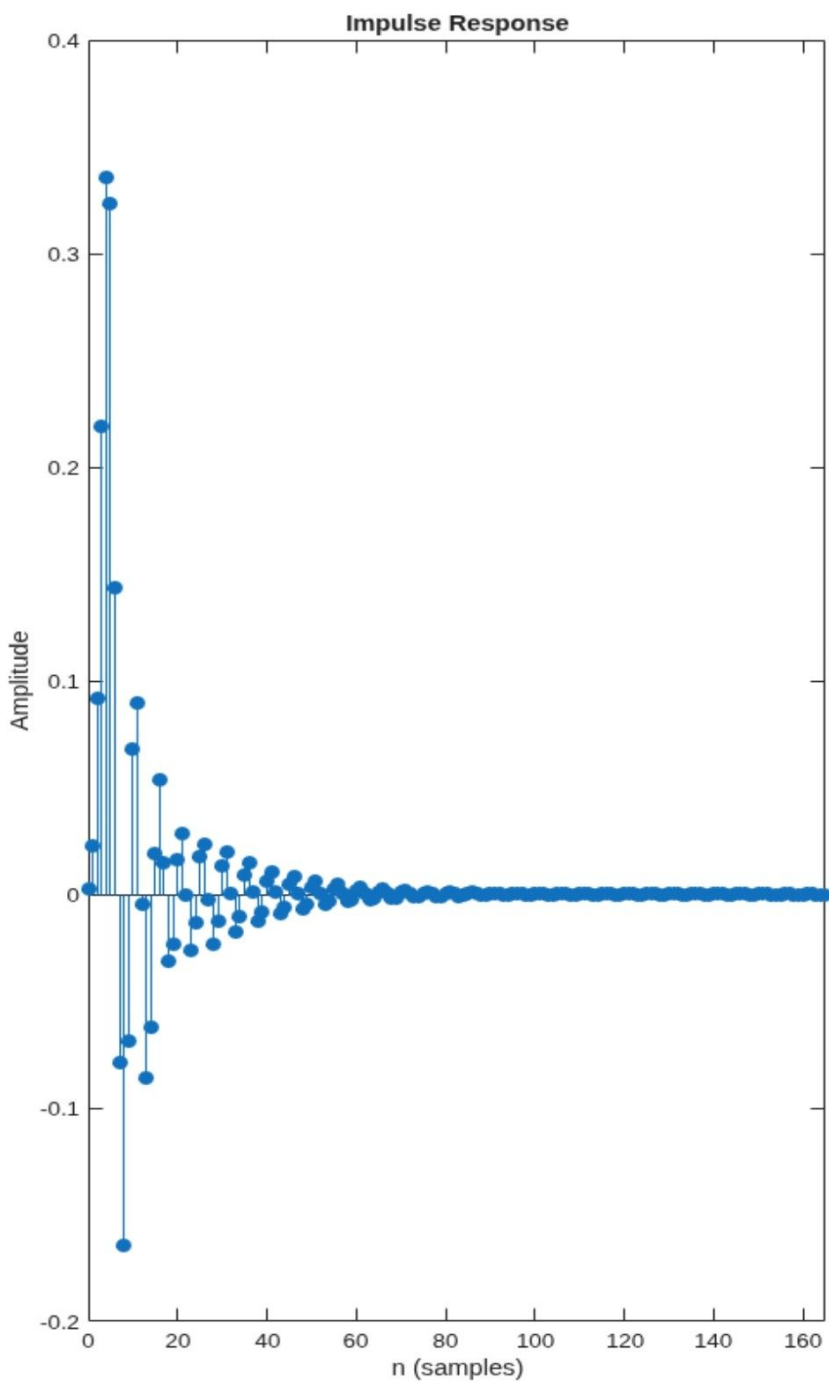


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Figure 2: Dsp2_1



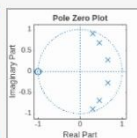
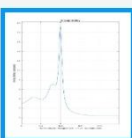
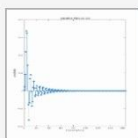
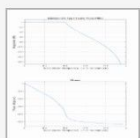
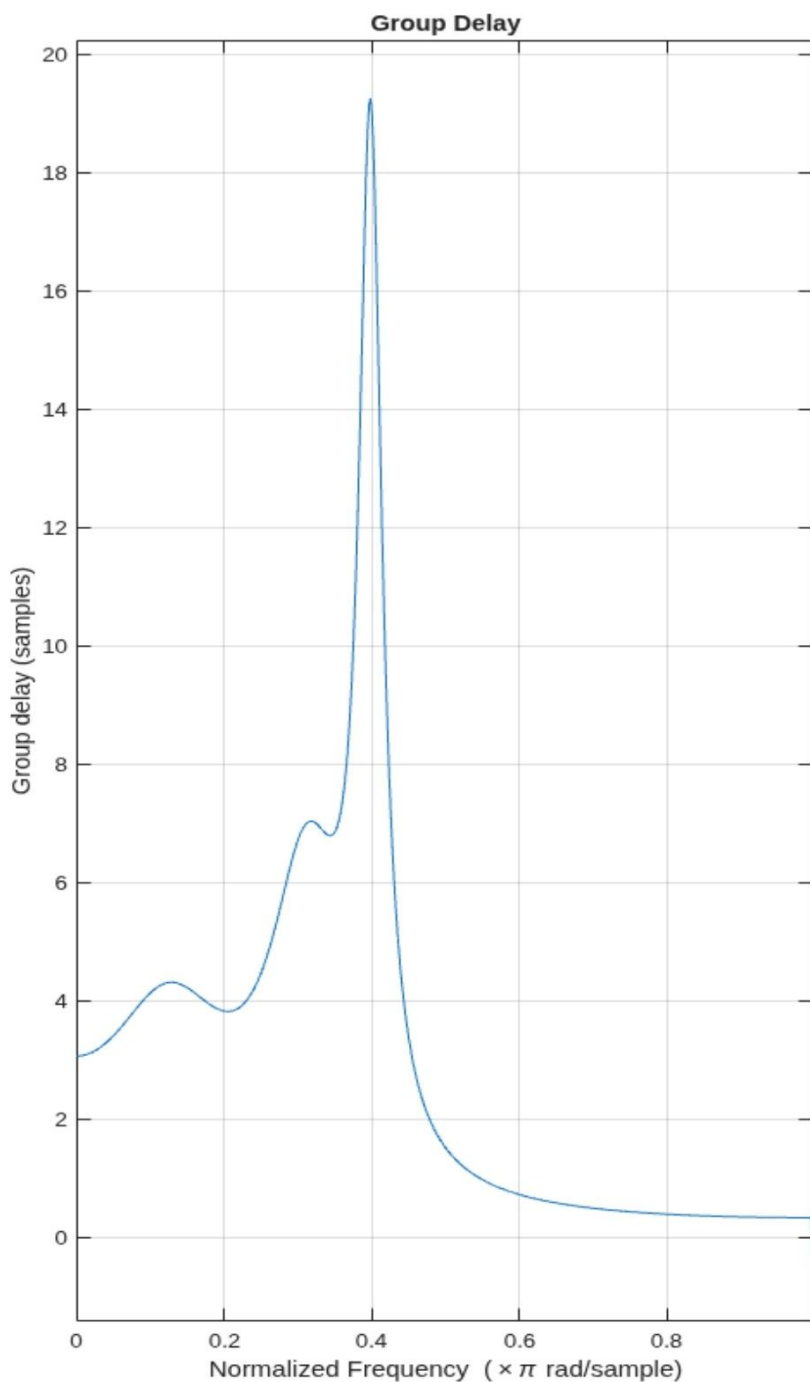


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Figure 3: Dsp2_1



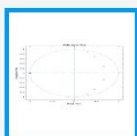
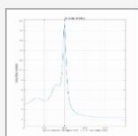
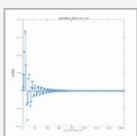
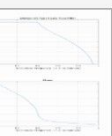
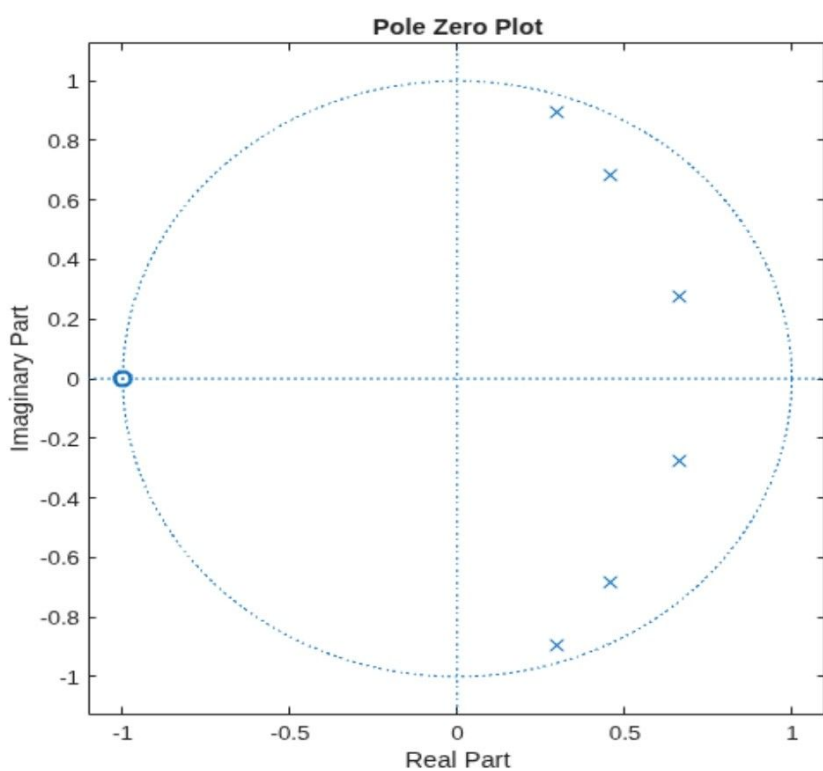


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Figure 4: Dsp2_1





MATLAB Drive > dsp2_2.m

```
1 clc;
2 clear;
3 close all;
4
5 % Specifications
6 Rp = 1;
7 Rs = 40;
8
9 Wp = 0.5;
10 Ws = 0.35;
11
12 % Filter Order
13 [N,Wn] = cheblord(Wp,Ws,Rp,Rs);
14
15 % High Pass Design
16 [b,a] = cheby1(N,Rp,Wn,'high');
17
18 fprintf('Filter Order = %d\n',N);
19 fprintf('Cutoff Frequency = %.4f\n',Wn);
20
21 % Stability Verification
22 poles = roots(a);
23
24 if all(abs(poles)<1)
25     disp('System is STABLE');
26 else
27     disp('System is UNSTABLE');
28 end
29
30
31 % Frequency Response
32 figure;
33 freqz(b,a);
34 title('Chebyshev Type-I High Pass Filter');
35
36 % Impulse Response
37 figure;
38 impz(b,a);
39 title('Impulse Response');
40
41 % Group Delay
42 figure;
43 grpdelay(b,a);
44 title('Group Delay');
45
46 % Pole Zero Plot
47 figure;
```



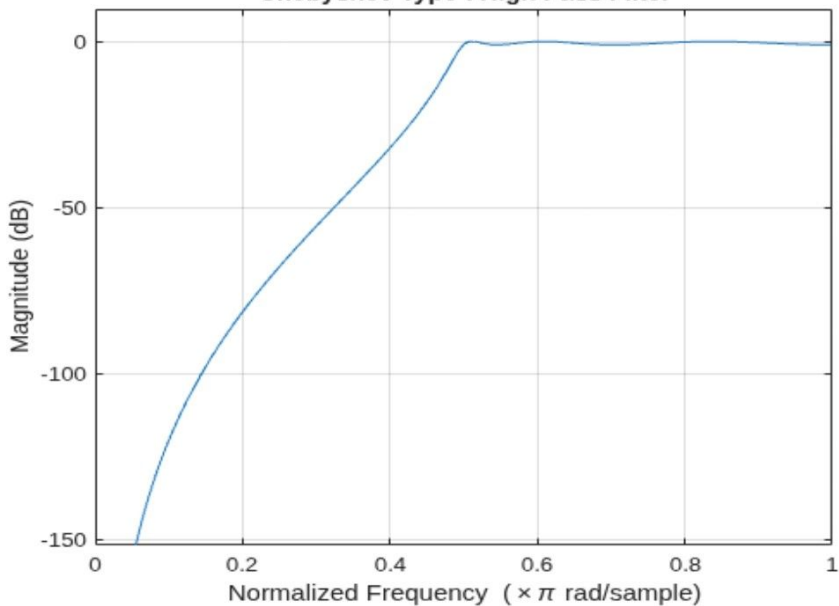
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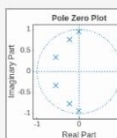
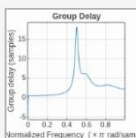
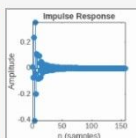
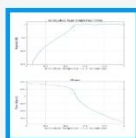
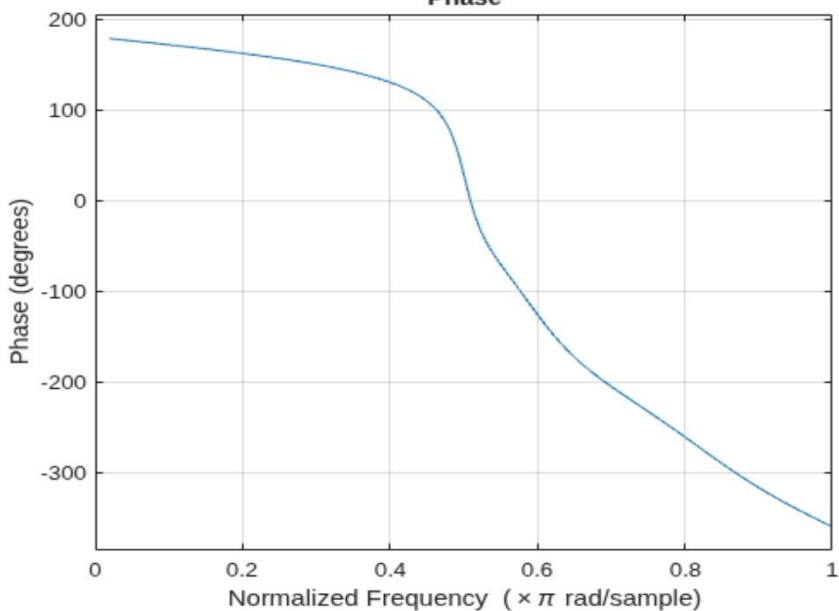


Figure 1: dsp2_2

Chebyshev Type-I High Pass Filter



Phase



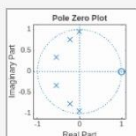
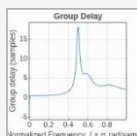
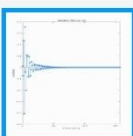
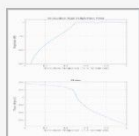
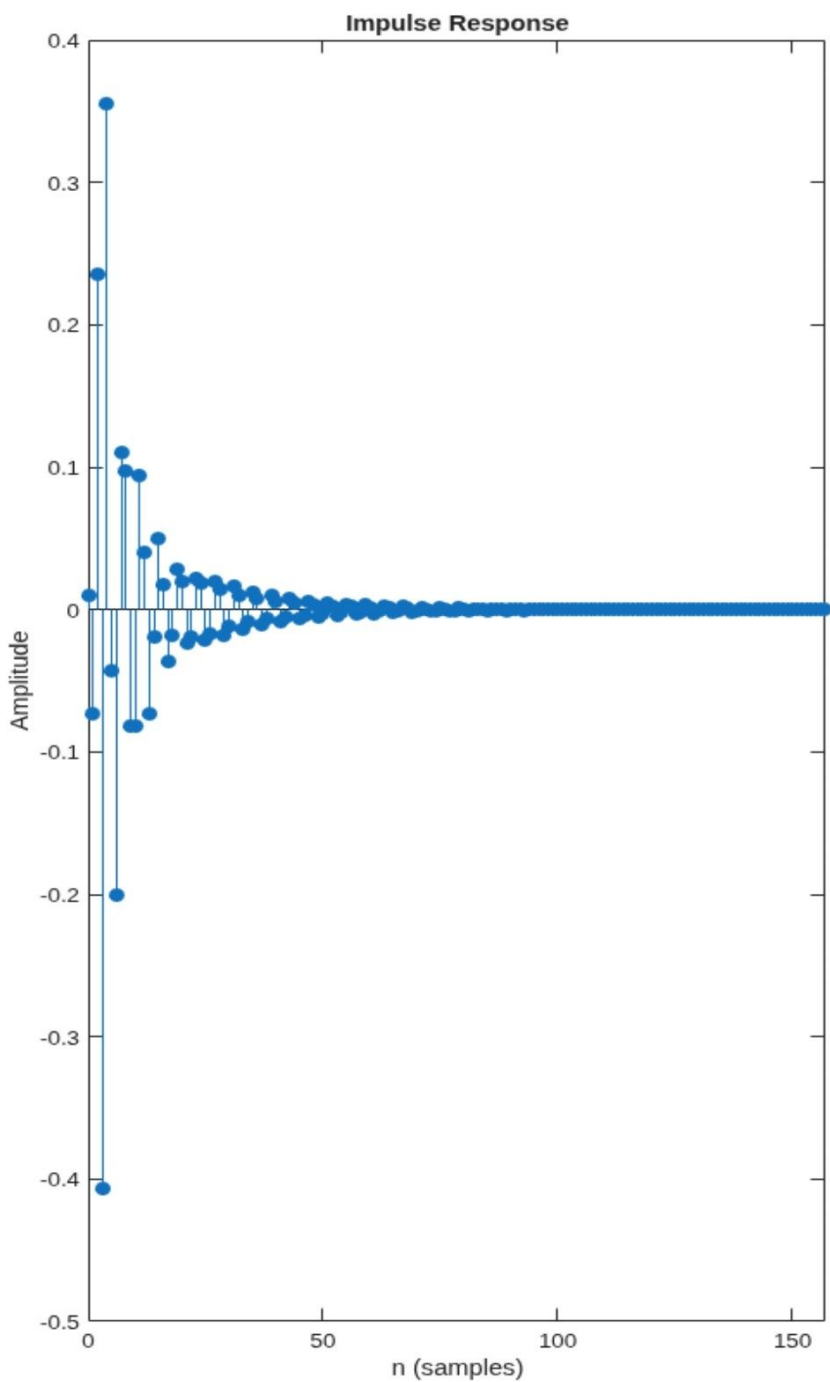


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Figure 2: dsp2_2



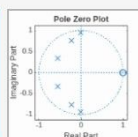
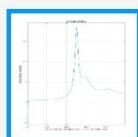
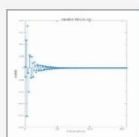
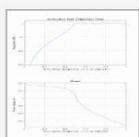
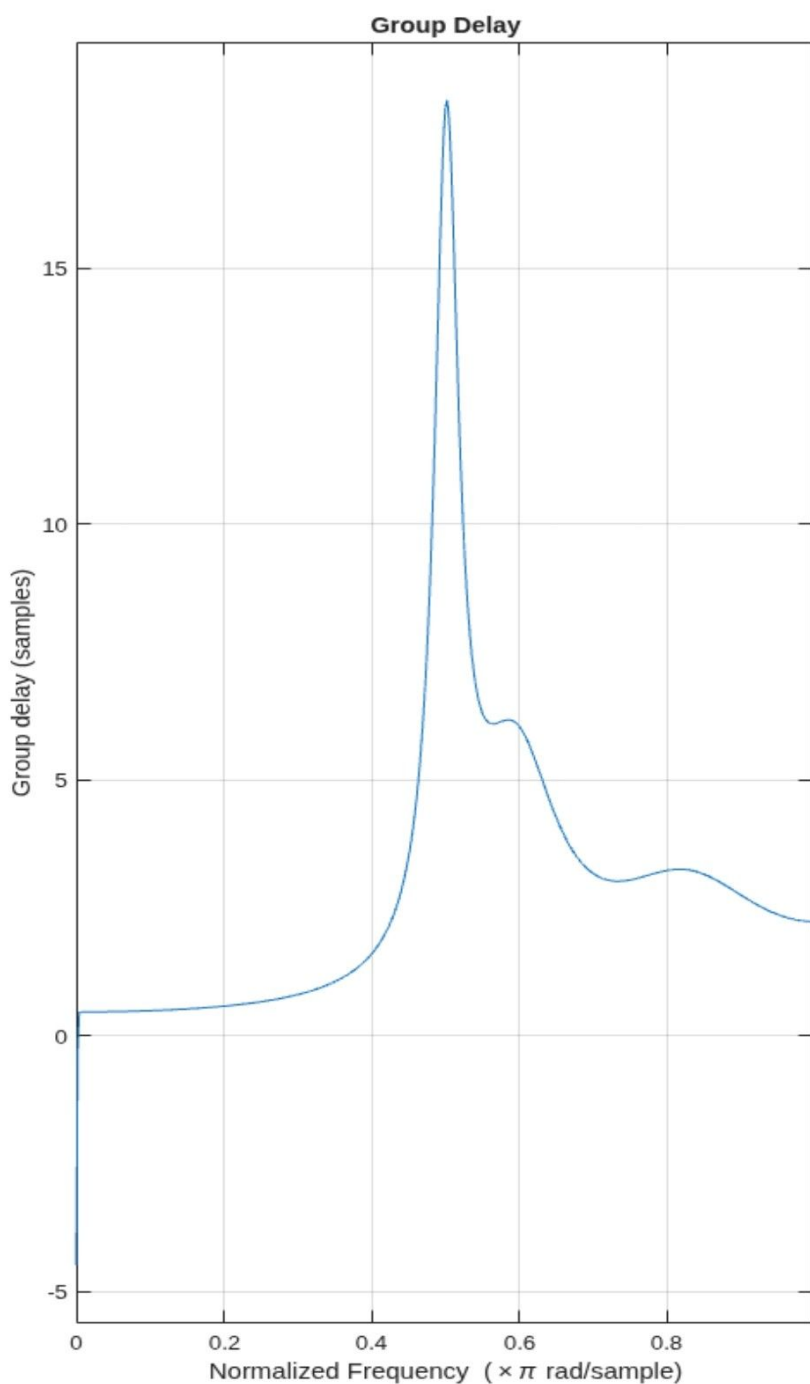


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Figure 3: dsp2_2





Figures

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Figure 4: dsp2_2

